
Quant Finance Study Plan

Module 1: Calculus

- **Timeline:** 2 Weeks
- **Subtopics:**
 - Limits, Continuity, Differentiability
 - Ordinary and Partial Derivatives
 - Chain Rule, Product Rule, Quotient Rule
 - Taylor Series Expansions
 - Optimization (finding maxima/minima)
 - Integration (definite and indefinite)
 - Fundamental Theorem of Calculus
 - *Math Evolution:* From the foundations laid by Newton and Leibniz to the development of advanced concepts like the Taylor series.
- **Resources:**
 - **Book:** "Calculus" by James Stewart (early chapters)
 - **Book:** "Mathematical Methods for Physics and Engineering" by Riley, Hobson, Bence (selected chapters on relevant calculus topics)
 - **Online:** Khan Academy Calculus series, MIT OpenCourseware Calculus courses
- **Evaluation Methods:**
 - Conceptual quizzes on each subtopic.
 - Problem sets involving calculations and proofs related to derivatives, integrals, and optimization.

- Derivation of key formulas (e.g., Taylor series, integration by parts).
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Module 2: Linear Algebra

- **Timeline:** 2 Weeks
 - **Subtopics:**
 - Vectors and Matrices
 - Matrix Operations (addition, multiplication, inverse, transpose)
 - Determinants
 - Eigenvalues and Eigenvectors
 - Linear Transformations
 - Systems of Linear Equations
 - *Math Evolution:* From solving systems of equations to the development of abstract vector spaces and spectral theorems.
 - **Resources:**
 - **Book:** "Introduction to Linear Algebra" by Gilbert Strang
 - **Book:** "Linear Algebra Done Right" by Sheldon Axler (for a more theoretical approach)
 - **Online:** Khan Academy Linear Algebra series, MIT OpenCourseware Linear Algebra courses, 3Blue1Brown's Essence of Linear Algebra series
 - **Evaluation Methods:**
 - Conceptual quizzes on each subtopic.
 - Problem sets involving matrix operations, finding eigenvalues and eigenvectors, and solving systems of linear equations.
 - Proofs of basic linear algebra theorems.
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Module 3: Probability Theory

- **Timeline:** 2 Weeks
- **Subtopics:**
 - Basic Probability (sample spaces, events, axioms)
 - Conditional Probability and Independence
 - Bayes' Theorem
 - Random Variables (discrete and continuous)
 - Probability Distributions (Bernoulli, Binomial, Poisson, Normal, Exponential, etc.)
 - Expectation, Variance, Standard Deviation, Covariance, Correlation
 - Central Limit Theorem
 - *Math Evolution:* From early work on games of chance to the rigorous axiomatization by Kolmogorov and the development of key distributions.
- **Resources:**
 - **Book:** "A First Course in Probability" by Sheldon Ross
 - **Book:** "Introduction to Probability" by Dimitri P. Bertsekas and John N. Tsitsiklis
 - **Online:** Khan Academy Probability series, MIT OpenCourseware Probability courses
- **Evaluation Methods:**
 - Conceptual quizzes on each subtopic.
 - Problem sets involving calculating probabilities, expectations, variances, and applying Bayes' Theorem.
 - Problems related to common probability distributions and the Central Limit Theorem.
 - **Green Book:** Brainstellar Interview Question

Module 4: Statistics

- **Timeline:** 2 Weeks
- **Subtopics:**
 - Descriptive Statistics (mean, median, mode, variance, etc.)
 - Inferential Statistics (hypothesis testing, confidence intervals)
 - Regression Analysis (linear and multiple regression, model fitting, diagnostics)
 - Time Series Analysis (AR, MA, ARMA models, stationarity)
 - *Math Evolution:* From descriptive measures to statistical inference and time series analysis.
- **Resources:**
 - **Book:** "OpenIntro Statistics" (free online resource)
 - **Book:** "The Elements of Statistical Learning" by Hastie, Tibshirani, and Friedman (for a more advanced treatment)
 - **Online:** Khan Academy Statistics series, various online tutorials on regression and time series analysis
- **Evaluation Methods:**
 - Conceptual quizzes on each subtopic.
 - Problem sets involving hypothesis testing, confidence intervals, and regression analysis.
 - Analysis of sample financial data using statistical techniques.

Module 5: Financial Markets and Instruments

- **Timeline:** 2 Weeks

- **Subtopics:**
 - Stocks, Bonds, Options, Futures, Swaps, Forwards
 - Market Microstructure (order books, bid-ask spreads, market makers)
 - Trading and Settlement
 - **Resources:**
 - **Book:** "Options, Futures, and Other Derivatives" by John C. Hull (selected chapters)
 - **Book:** "Trading and Exchanges: Market Microstructure for Practitioners" by Larry Harris
 - **Online:** Investopedia, financial news websites (e.g., Bloomberg, Wall Street Journal)
 - **Evaluation Methods:**
 - Conceptual quizzes on different financial instruments and market microstructure.
 - Discussion of how different instruments are traded and their characteristics.
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Module 6: Time Value of Money

- **Timeline:** 1 Week
- **Subtopics:**
 - Discounting and Compounding
 - Present Value and Future Value
 - Annuities and Perpetuities
- **Resources:**
 - **Book:** "Options, Futures, and Other Derivatives" by John C. Hull (relevant chapters)

- **Online:** Investopedia, Khan Academy (Finance and Capital Markets section)
 - **Evaluation Methods:**
 - Conceptual quizzes on discounting, compounding, and different types of cash flows.
 - Problem sets involving calculating present and future values, annuity payments, etc.
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Module 7: Portfolio Theory

- **Timeline:** 2 Weeks
- **Subtopics:**
 - Risk and Return
 - Diversification
 - Mean-Variance Optimization
 - Efficient Frontier
 - Capital Asset Pricing Model (CAPM)
 - *Math Evolution:* From Markowitz's portfolio optimization to the CAPM.
- **Resources:**
 - **Book:** "Investments" by Bodie, Kane, and Marcus (selected chapters)
 - **Paper:** "Portfolio Selection" by Harry Markowitz (original paper)
- **Evaluation Methods:**
 - Conceptual quizzes on risk, return, diversification, and the CAPM.
 - Problem sets involving calculating portfolio risk and return, constructing efficient portfolios.
 - Derivation of the CAPM formula.

Module 8: Options Pricing

- **Timeline:** 4 Weeks
- **Subtopics:**
 - Option Payoffs and Strategies
 - Binomial Option Pricing Model
 - Black-Scholes-Merton Model
 - Stochastic Calculus (Brownian Motion, Ito's Lemma)
 - Partial Differential Equations (Black-Scholes PDE)
 - Option Greeks (Delta, Gamma, Vega, Theta, Rho)
 - Volatility (historical, implied)
 - *Math Evolution:* From early option pricing to Black-Scholes-Merton and the application of stochastic calculus.
- **Resources:**
 - **Book:** "Options, Futures, and Other Derivatives" by John C. Hull
 - **Book:** "Paul Wilmott on Quantitative Finance" (selected chapters)
 - **Paper:** "The Pricing of Options and Corporate Liabilities" by Black and Scholes (original paper)
 - **Online Resources:** Quantstart, Quantopian
- **Evaluation Methods:**
 - Conceptual quizzes on option payoffs, strategies, and the Black-Scholes model.
 - Problem sets involving pricing options using the binomial and Black-Scholes models.
 - Derivation of the Black-Scholes PDE and understanding its solution.
 - Calculating and interpreting option Greeks.

- Coding exercises to implement option pricing models in Python.
 - **Green Book:** Questions from Option Pricing and its assumptions.
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Module 9: Fixed Income

- **Timeline:** 3 Weeks
 - **Subtopics:**
 - Bond Valuation
 - Yield Curve
 - Duration and Convexity
 - Interest Rate Models (Vasicek, CIR)
 - *Math Evolution:* From basic bond pricing to advanced interest rate models.
 - **Resources:**
 - **Book:** "Options, Futures, and Other Derivatives" by John C. Hull (relevant chapters)
 - **Book:** "Fixed Income Securities" by Veronesi
 - **Evaluation Methods:**
 - Conceptual quizzes on bond valuation, yield curves, and interest rate risk.
 - Problem sets involving calculating bond prices, yields, duration, and convexity.
 - Understanding and applying interest rate models.
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Module 10: Risk Management

- **Timeline:** 2 Weeks
- **Subtopics:**
 - Value at Risk (VaR)

- Expected Shortfall (ES)
 - Credit Risk
 - Operational Risk
 - **Resources:**
 - **Book:** "Risk Management and Financial Institutions" by John C. Hull
 - **Book:** "Value at Risk: The New Benchmark for Managing Financial Risk" by Philippe Jorion
 - **Evaluation Methods:**
 - Conceptual quizzes on different types of financial risk.
 - Problem sets involving calculating VaR and ES.
 - Discussion of credit risk and operational risk management techniques.
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Module 11: Advanced Topics (Select 2-3 based on interest)

- **Timeline:** 4 Weeks (2-3 weeks per topic)
- **Topics:**
 - **Stochastic Processes beyond Brownian Motion:**
 - Poisson Processes
 - Jump Diffusion
 - Levy Processes
 - **Advanced Option Pricing Models:**
 - Local Volatility Models
 - Stochastic Volatility Models (Heston, SABR)
 - **Numerical Methods:**
 - Monte Carlo Simulation
 - Finite Difference Methods
 - **Machine Learning in Finance:**

- Supervised Learning (Regression, Classification)
 - Unsupervised Learning (Clustering, Dimensionality Reduction)
 - Reinforcement Learning
 - **Algorithmic Trading and Market Making:**
 - High-Frequency Trading
 - Statistical Arbitrage
 - **Econometrics:**
 - Advanced time-series models (GARCH)
 - Panel data analysis
 - **Resources:** (Varies depending on the chosen topics)
 - Advanced textbooks and research papers in the selected areas.
 - **Evaluation Methods:**
 - In-depth problem sets and coding exercises.
 - Potentially a small project involving implementing a model or algorithm.
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Notes and Questions

- (Prepare your own notes, questions, and reflections throughout the study process)
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Important Considerations:

- **Green Book:**
 - Heard on the Street
 - Quant Job Interview Questions And Answers - Mark Joshi
- **Evaluation:**

- For probability and Option Pricing, do a lot of questions from the green book and brainstellar. These two are very important to get hands-on practice for interviews.